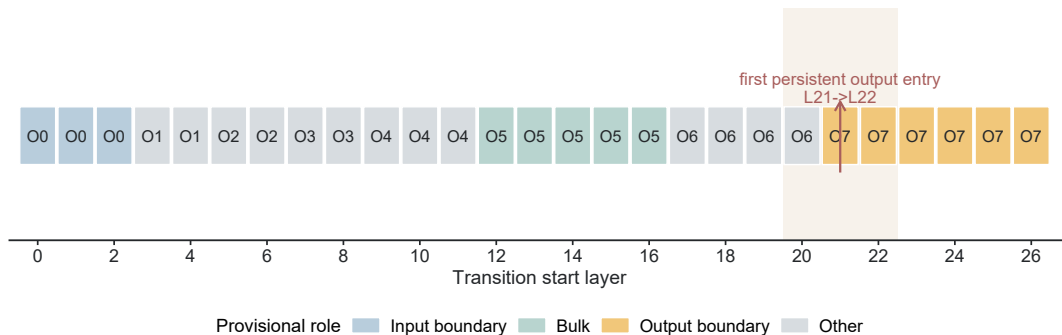


Operator clustering revealed a late macro transition, not universal primitives

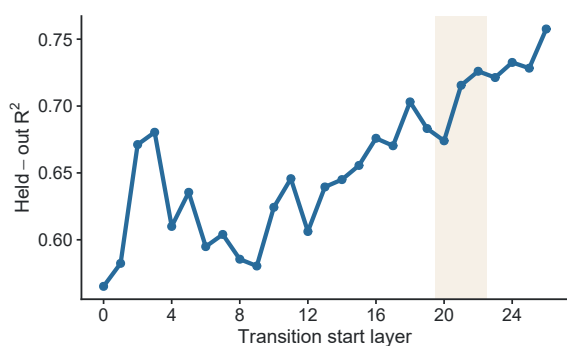
a Initial clustering suggested boundary-bulk-boundary organization

Operator IDs show the fine K=8 staircase; role labels were provisional



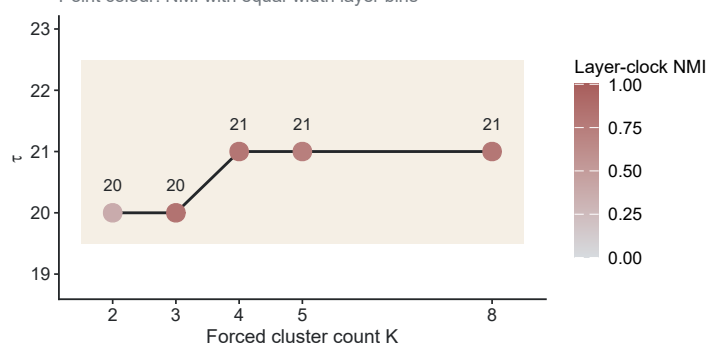
Provisional role Input boundary Bulk Output boundary Other

b Adjacent-layer operators remain measurable

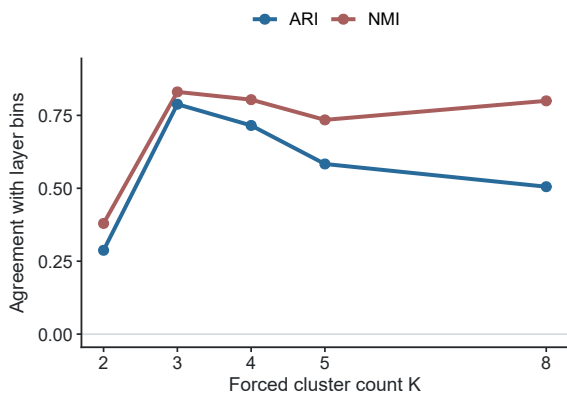


c The late macro transition survives forced K

Point colour: NMI with equal-width layer bins



d Fine labels contain substantial layer-clock signal



e Removing smooth depth trends destabilizes fine boundaries

Shaded band: retained L20-L22 macro zone

